Sajid Ali

cs25mtech11019@iith.ac.in | bhatsajid425@gmail.com | +91 9682175746 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.38
Government Boys Higher Secondary School, Safapora, Ganderbal	Class XII (Jammu & Kashmir Board of School Education)	2020	89.60%
Government Boys High School, Chewa, Bandipora	Class X (Jammu & Kashmir Board of School Education)	2018	85.00%

POSITIONS OF RESPONSIBILITY

Teaching Assistant | IIT Hyderabad

- **Introduction to Programming** (August 2025 - November 2025)
- **Compilers I** (March 2026 - April 2026)
- **Foundations of Machine Learning** (August 2026 - November 2026)

PROJECTS

Uncertainty Estimation in Deep Learning Models | Guide: Prof. Srijith P. K.

- Design a **knowledge distillation framework in PyTorch** in which a single student network predicts both **class probabilities and predictive uncertainty** for deep learning classifiers.
- Leverage **out-of-distribution (OOD) data** during distillation, enabling the student to separate in-distribution from OOD inputs and learn **uncertainty-aware representations**.
- Decompose **epistemic and aleatoric uncertainty** by transferring knowledge from an **ensemble of teacher models** into one efficient student model.

TransactiWar - Secure Banking Web Application | Network Security | GitHub

- Co-developed a framework-free money-transfer web application in **PHP and PostgreSQL**, implementing **authentication, session management, Cross-Site Request Forgery (CSRF) protection**, rate limiting and audit logging.
- Prevented **race conditions and brute-force attacks** using **PostgreSQL row-level locking (SELECT ... FOR UPDATE)** and atomic upsert-based rate limiting.
- Containerized the application with **Docker** behind **HTTPS-only Apache (TLS)**, then defended it against peer teams' **SQL injection, Cross-Site Scripting (XSS), CSRF and IDOR** attacks in a live **Capture-the-Flag (CTF)** war game.

Business Entity Resolution | Amazon ML Challenge 2026 | GitHub

- Engineered a **two-stage entity resolution pipeline in Python** that links noisy, **multilingual (English and Indic-script)** business records across three sources with no common identifiers.
- Fine-tuned **LaBSE with Sentence-Transformers (PyTorch, Hugging Face)** using **contrastive learning and hard-negative mining**, powering **CUDA-accelerated nearest-neighbor retrieval**.
- Trained a **LightGBM gradient-boosting classifier** on **RapidFuzz fuzzy-matching, scikit-learn TF-IDF and word-rarity features**, tuning decision thresholds for **high precision** and generalization to an **unseen country**.

Binary Image Classification with SE-ResNet | Deep Learning | GitHub

- Implemented a custom **SE-ResNet convolutional neural network (CNN) from scratch in PyTorch**, combining **residual connections** with **squeeze-and-excitation channel attention** and label-smoothed binary cross-entropy loss.
- Improved robustness and generalization through **cosine-annealed Adam**, **test-time augmentation (TTA)** and **snapshot ensembling** for the IIT Hyderabad Deep Learning Hackathon.

RELEVANT COURSES

AI and Machine Learning: Foundations of Machine Learning, Deep Learning

Systems and Networking: Advanced Computer Networks, Networked Wireless Systems, Network Security

Theory and others: Advanced DSA, Probability Theory, DBMS, OS

SKILLS

Programming Languages: C, C++, JavaScript, Python

Machine Learning: Numpy, Pandas, Sklearn, PyTorch, Matplotlib, Seaborn, Hugging Face

Deep Learning: CNN, RNN, LSTM, GRU, Transformer, Autoencoder

Retrieval & AI Systems: RAG, Dense Retrieval, BM25, Hybrid Retrieval, RRF, FAISS, BGE Embeddings, Cross-Encoder Reranking

Web Tech & Database: HTML, CSS, React.js, SQL

Networking Tools: Wireshark, TCP dump, ns-3

Core Competencies: Problem-Solving, Effective-Communication, Teamwork, Adaptability, Creativity

ACHIEVEMENTS

- Solved over 600 problems across LeetCode and geeksforgeeks

EXTRACURRICULAR

- Active participant in the Clean India drive at IIT Hyderabad
- Play table tennis and cricket regularly, which keeps me active and builds teamwork and discipline